SCHOOL OF COMPUTING & DATA SCIENCE
The University of Hong Kong
AI & Data Science

School of Computing and Data Science
Department of AI and Data Science • The University of Hong Kong

PRINCIPAL INVESTIGATOR
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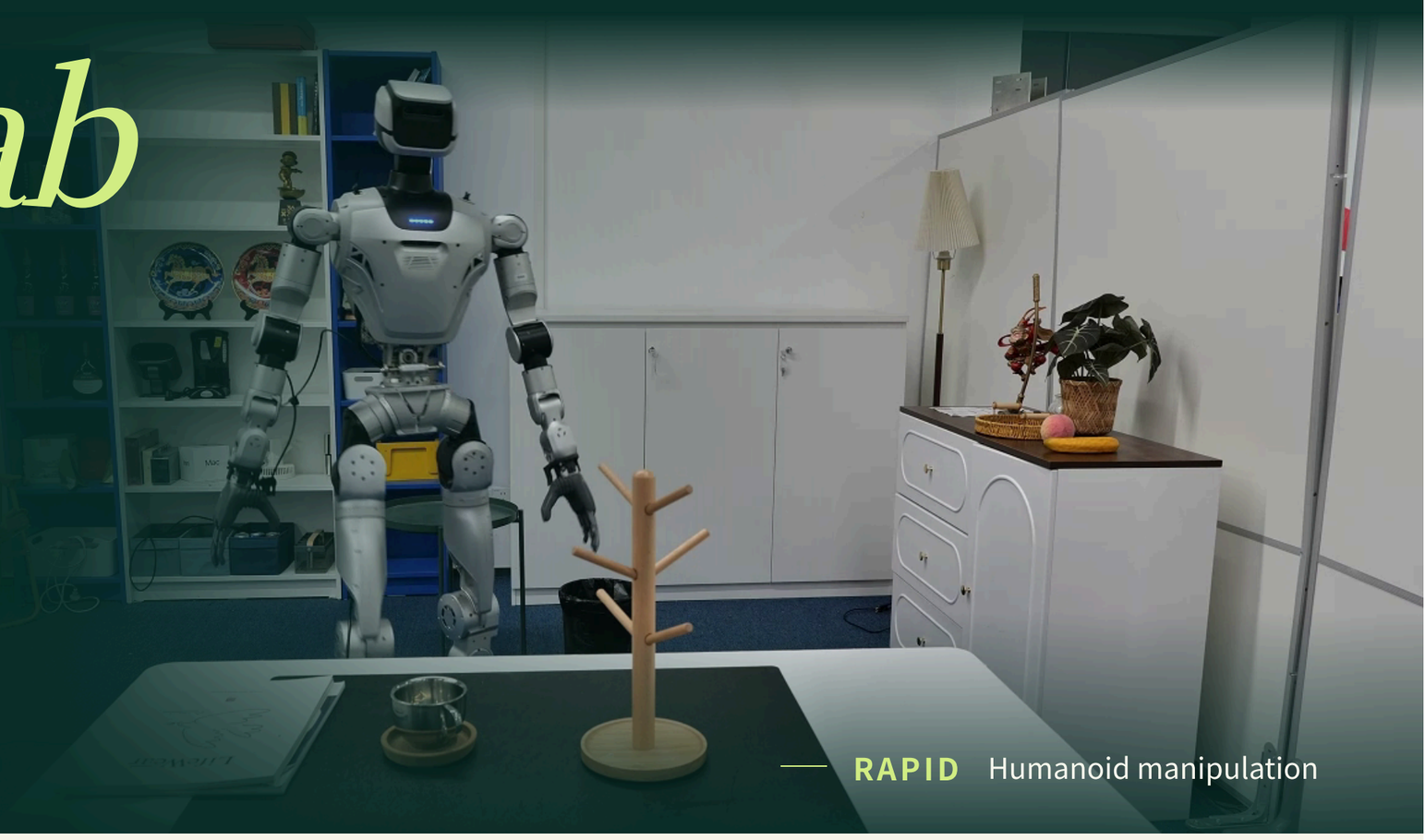


PANDA *Lab*

Physical AI and Autonomous Robotics

Embodied AI for the real world

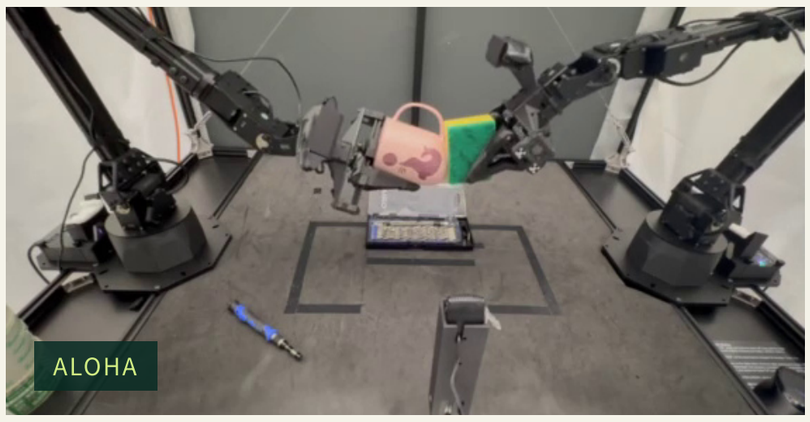
We unite perception, learning, planning and control to advance robot manipulation and navigation, building reliable, generalizable physical intelligence through experiments on real robots.



RAPID Humanoid manipulation

01 From perception to action

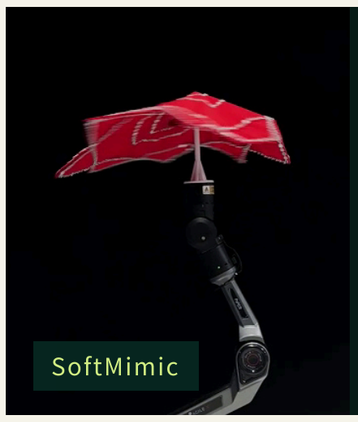
RESEARCH IN THE REAL WORLD



ALOHA



GauSS-MI



SoftMimic

Understand physics. Learn to act.

DEFORMABLE MANIPULATION

Learning & manipulation

Imitation learning, bimanual dexterity and long-horizon task planning.

Perception & navigation

Active 3D reconstruction, scene understanding and motion planning.

Physical interaction

Dynamics learning, tactile sensing and skill transfer for deformable objects.

Robot platforms

EXPERIMENTAL SYSTEMS

UR • **Franka** • **ALOHA** • **G1** • **LimX Oli** • **Tron**

Robotic arms, bimanual systems, humanoids and mobile platforms for real-world experiments.

02 Selected publications

RESEARCH HIGHLIGHTS • 2024-2026

Nature Machine Intelligence 2024

Soft skin with decoupled force sensing

A soft skin with self-decoupled three-axis force-sensing taxels

Y. Yan, A. Zermane, **J. Pan** & A. Kheddar • 6, 1284-1295

Nature Communications 2024

Robust vision in extreme light

Embodied neuromorphic synergy for lighting-robust machine vision to see in extreme bright

S. Lin et al., including **J. Pan** • 15, 10781

IJRR 2026

BIM-Loc: Prior-guided indoor localization

BIM-Loc: BIM-integrated discrepancy-aware LiDAR-based indoor localization

Y. Zhang et al., including **J. Pan** • The International Journal of Robotics Research

IEEE Transactions on Robotics 2025

NeuPAN: Model-based robot navigation

NeuPAN: Direct Point Robot Navigation with End-to-End Model-Based Learning

R. Han et al., including **J. Pan** • 41, 2804-2824

Perceive the world. Learn to act.

PANDA Lab • The University of Hong Kong

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Explore our research

panda-lab-hku.github.io

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